ESPRIT École d'Ingénieurs

INTERNSHIP JOURNAL

Engineering Internship | | Min. 8 weeks

GENERAL INFORMATION

Student name	**Farah AYARI**	
Level / Year	4th year → 5th year	
Student ID		
Host Organization	Name : **Al Baraka Bank Tunisia**	
	Adress : Av. Cheikh Mohamed Fadhel Ben Achour, Centre Urbain Nord, 1082 Tunis	
Company Supervisor	Name : **Mrs. Ahlem Ben Haddoud**	
	Mail :	
	Tel :	
Internship Period	From : **03/08/2026**	To : **03/10/2026**
Duration	8 weeks minimum	

Supervisor Stamp & Signature

INTERNSHIP JOURNAL (to be completed by the intern)

Date	Tasks Completed
03/08/2026	Onboarding at Al Baraka head office and project scope definition.
04/08/2026	Study of Banking Law n° 2016-48 and BCT Circular n° 2019-08.
05/08/2026	Analysis of the internal Islamic banking guide and training primer.
06/08/2026	Definition of the seven project objectives and technical constraints.
07/08/2026	Review of RAG failure points and selection of the technology stack.
10/08/2026	Setup of the RAGLab repository, offline tests, and CI workflows.
11/08/2026	Implementation of PDF/DOCX loaders and Arabic text normalization.
12/08/2026	Development of fixed-window token chunking and ChromaDB indexing.
14/08/2026	Integration of NVIDIA Nemotron embeddings and LLM API clients.
17/08/2026	Creation of the frozen 50-question trilingual evaluation benchmark.
18/08/2026	Implementation of grounded JSON prompting and quote verification.
19/08/2026	Reverse-engineered specification and banking governance gap analysis.
20/08/2026	Corpus reset on the four real documents and step-and-plan protocol.
21/08/2026	Linguistic repair of Law n° 2016-48 restoring 198 article headings.
24/08/2026	Linguistic repair of Circular n° 2019-08, internal guide, and primer.
26/08/2026	Implementation of structure-aware chunking with heading breadcrumbs.
27/08/2026	Addition of index fingerprints, SHA-256 store, and ingest seal.
28/08/2026	Paired live chunking benchmark: hit@1 improved from 51% to 71%.
31/08/2026	Live comparison of dense vs. hybrid retrieval; dense kept default.
01/09/2026	Extraction of 198 knowledge units from Law n° 2016-48 with checks.
02/09/2026	Construction of 77 cross-references and 53 structured number records.

Date	Tasks Completed
03/09/2026	Addition of six governance axes and dedicated knowledge-unit index.
04/09/2026	Design of the 10-question Arabic target set for complex queries.
07/09/2026	Implementation of query intent routing and evidence sufficiency check.
08/09/2026	Hardening of the citation gate and six-reason refusal taxonomy.
09/09/2026	Addition of PII output scrubbing, JSONL audit logs, and safety tests.
10/09/2026	Answer-model A/B benchmark confirming 1.000 cited-source overlap.
11/09/2026	Freezing of the 27-route FastAPI contract with token authentication.
14/09/2026	Design of the trust boundary between Bank Assistant and RAGLab.
15/09/2026	PostgreSQL 16 schema setup with Flyway and client/staff separation.
16/09/2026	Implementation of JWT auth, refresh rotation, and 9-permission RBAC.
17/09/2026	Two-phase chat quota metering under 64 striped session locks.
18/09/2026	Ephemeral customer context masking to protect personal banking data.
21/09/2026	Development of the Angular 21 trilingual portal with RTL layout.
22/09/2026	Implementation of read-only account views and financing simulator.
23/09/2026	Edge answer sanitizer and bank refusal wording for client privacy.
24/09/2026	Implementation of verified SSE chat streaming via Nginx proxy.
25/09/2026	Development of the French staff console for supervision and RAGLab.
28/09/2026	Security review against OWASP LLM Top 10 and HTTP security headers.
29/09/2026	Containerization with Docker Compose, healthchecks, and metrics.
30/09/2026	Fixes for live defects: duplicate-ingest guard and history scrubbing.
01–02/10/2026	Full end-to-end regression testing and CI verification on all suites.
03/10/2026	Final handover package and internship report review with supervisor.

INTERNSHIP EVALUATION (to be completed by the company supervisor)

1 – Unsatisfactory	Tasks do not meet required expectations.
2 – Satisfactory	Tasks meet required expectations.
3 – Very Satisfactory	Tasks exceed required expectations.
4 – Excellent	Tasks significantly exceed expectations and merit special recognition.

CRITERIAL EVALUATION GRID

Evaluation Criteria	Unsatisfactory (1)	Satisfactory (2)	Very Satisfactory (3)	Excellent (4)	Score (1-4)
A. Behavior & Professional Attitude					
1. Motivation and engagement	Manifest disinterest, little involvement	Limited motivation, passive participation	Good motivation, active participation	Remarkable enthusiasm, very high involvement	
2. Initiative and drive	Systematically waits for instructions	Rarely takes initiative	Proposes ideas and acts without being asked	Very proactive, formulates innovative proposals	
3. Responsibility and deadline compliance	Does not meet deadlines, avoids responsibility	Deadlines partially met	Assumes responsibilities and meets deadlines	Model of rigor, anticipates and delivers early	
4. Adaptability	Great difficulty adapting to company practices	Slow and partial adaptation	Good adaptation to methods and culture	Rapid and exemplary integration	
5. Open-mindedness and social awareness	Closed to social realities, low empathy	Limited openness, partial awareness	Shows real openness to societal issues	High sensitivity, fine analysis of human issues	
6. Ethics and values	Behavior contrary to organizational values	Generally correct attitude, some lapses	Consistently ethical and respectful attitude	Role model for ethics and human values	
7. Teamwork and collaboration	Isolated, major relational difficulties	Participates but limited interactions	Collaborates well, contributes effectively	Unites the team, excellent collaborative dynamics	
8. Autonomous working ability	Constant need for supervision, no independent progress	Partial autonomy, frequent monitoring required	Works effectively autonomously	Full autonomy, independently handles complex situations	
B. Technical & Engineering Skills					
9. Technical mastery of specialty	Insufficient knowledge, many gaps	Basic knowledge, partial application	Good technical mastery of the specialty	Recognized and valued technical expertise	
10. Relevance of implemented solution	No solution or unsuitable solution	Solution proposed but insufficiently optimized	Correct, well-argued and adapted solution	Innovative, optimal, perfectly justified solution	
11. Use of resources, tools and methods	Does not use or misuses required tools	Basic use of tools	Good mastery of tools and methods	Expert mastery, fully exploits available tools	
Company Supervisor Score /20				**….. /20**	

Supervisor Stamp & Signature	**Remarks**

RECOMMENDATIONS FOR THE INTERN

DURING THE INTERNSHIP

The student engineer shall:

- Provide this journal to the company supervisor on the first day of the internship to define the internship program.
- Respect the employer's conditions: working hours, punctuality, discretion, discipline, etc.
- Accomplish the tasks defined in the internship program.

AFTER THE INTERNSHIP

The student engineer shall:

- At the start of the university year, the student must submit a digital version to the internship department:
- The internship completion certificate (original to be kept by the student).
- The internship report.
- This journal, duly completed and signed by the supervisor and the host organization (mandatory: completed manually, then scanned).

IMPORTANT NOTE

The student engineer shall:

- The submission link will be communicated to students by the internship department at the start of the academic year 25/26.